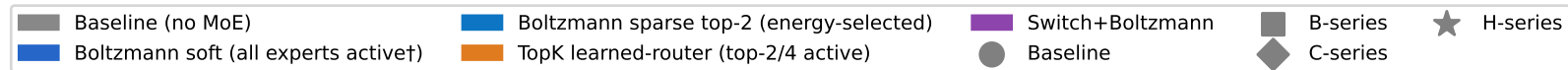
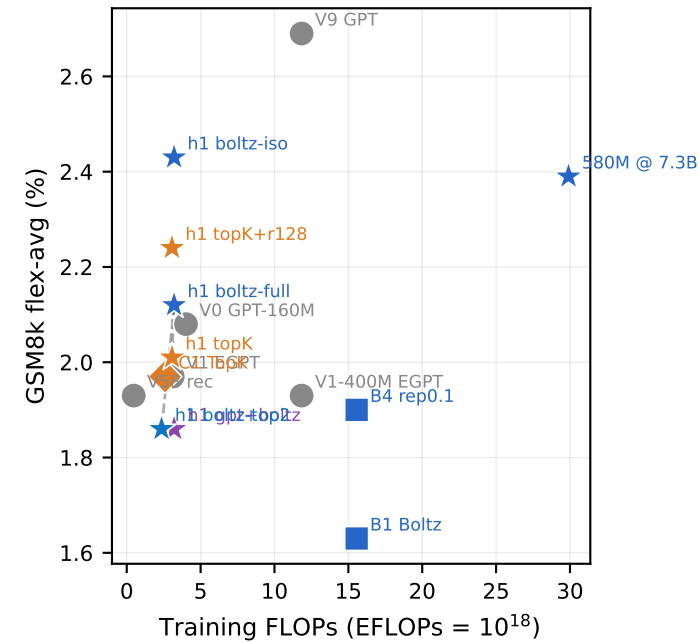
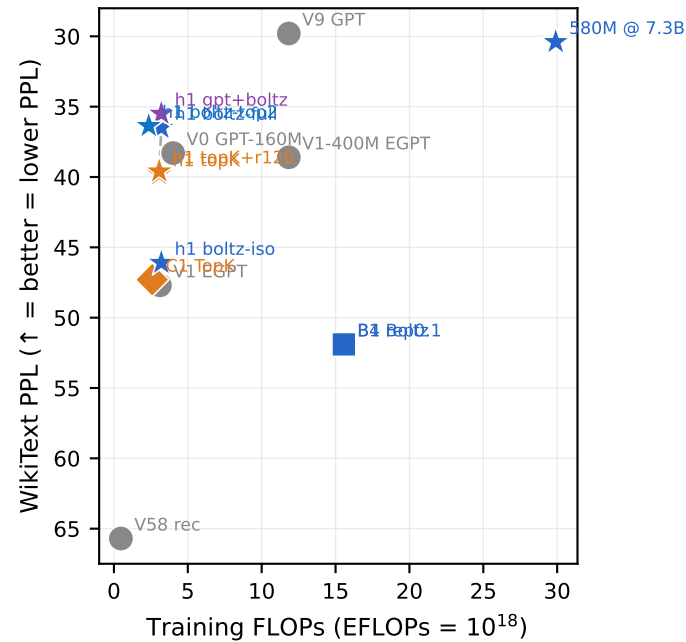
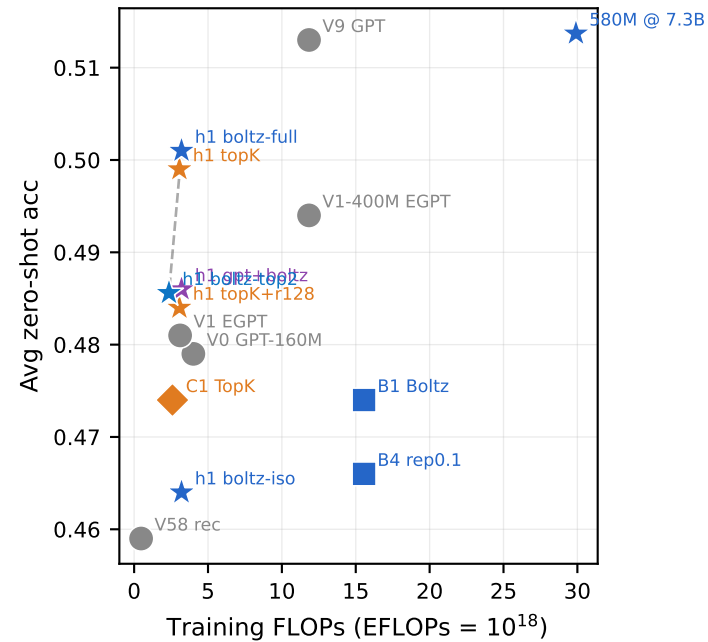


MoE variants vs baselines (7.86B tokens)
x-axis: training FLOPs $\approx 6 \cdot \text{active_params} \cdot \text{tokens}$



FLOPs = $6 \cdot \text{active_params} \cdot \text{tokens}$. Active params encode the sparsity assumption (Boltzmann soft = all experts active; sparse top-2 $\approx 75\%$ of soft).
Boltzmann routing itself is FLOPs-free vs the MoE forward — energy $E_i = x \cdot \text{term1}_i$ reuses term1 already needed for output.